www.absstem.com

absstem INDUSTRIAL

Quotation No : Q23000118

Date : 27-Mar-2023

Hi_Tech Power & Steel Limited
Vill-Parsada, PO-Sarora, Teh- Tilda
RAIPUR
CHATTISGARH

Kind Attention : Mr./Ms. Dilip Mishra / Harsh Kariwal/ Pankaj Tekriwal

Dear Sir,

We understand that in industry, reliability is not just a requirement,it is the foundation of success.
That is why Absstem Industrial PSA Generator is more than just a product; it is a commitment to excellence, efficiency, and uninterrupted performance.

Every component is designed with precision and innovation, ensuring that each molecule of oxygen and nitrogen is delivered with the utmost purity.

With fully automated operation, our system seamlessly integrates into your processes, reducing manual intervention and maximizing productivity.

Moreover, our industry-leading air ratio ensures that you achieve the highest efficiency, optimizing energy consumption while maintaining peak performance.
And because we believe in standing by our craftsmanship, we offer an unmatched-10 year warranty on sieves, giving you the peace of mind that your investment is protected for the long run.

We have attached the detailed quotation for your review.

Should you have any questions or require further customization to meet your specific needs, please do not hesitate to reach out.

We are here to provide not just a product but a solution that breathes life into your operations.

We look forward to the opportunity to serve you and build a lasting partnership.

Thanks and Regards,

For Absstem Industrial Solutions Pvt. Ltd.

Pramod Kumar

Director

8447642325

www.absstem.com

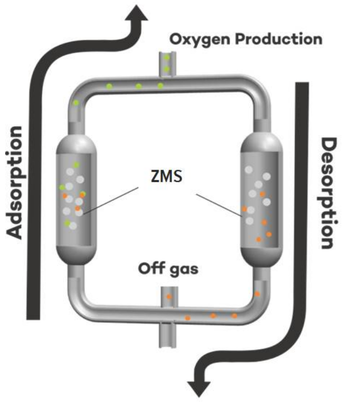
POWERED BY ABSSTEM INDUSTRIAL

Industrial

PSA Oxygen Generator

absstem
idos
absstem
PSA
Oxygen Generator

PSA Technology



PSA Process explaining oxygen production

Industrial Oxygen Generator mainly consists of two adsorbers filled with zeolite molecular sieves. Pressurized air is passed through one of these adsorbers, zeolite having a property to attract nitrogen, adsorbs Nitrogen from air and oxygen is allowed to pass through and move into the storage tank as the final product. As the adsorber approaches saturation, the air is allowed to pass through the other adsorber while this adsorber is being regenerated to ensure a continuous supply of Oxygen. This changeover takes place through a specially designed microcontroller to ensure a hassle-free operation. The quality of oxygen produced is monitored by an Oxygen analyser.

Key Features

- Shipped ready for installation
- 5 minutes Startup time with uninterrupted stoppages.
- Quick & Easy installation
- Low operating costs compared to gas cylinders/liquid oxygen
- Reduced Maintenance costs
- Embedded Safety Interlocks to cater any abnormality.
- Space-saving compact design.
- Eliminates all safety hazards which are there with gas cylinder or liquid oxygen.
- 10 years warranty on molecular sieves.

Description

a. Product	Oxygen Gas Generator
b. Model	**IDOS**
c. Technology	PSA (Pressure Swing Adsorption)

Specifications

a. Oxygen Flow Rate	As per Performance Sheet
b. Oxygen Purity	As per Performance Sheet
c. Oxygen Delivery Pressure	5.5 Barg (Higher Pressure Options Available)
d. Ambient conditions	
• Temperature	0 to 50 Deg. Celsius
• Relative Humidity	up to 80 % (Max.)
• Altitude	up to 1000 m above Sea level

Scheme Offered

This scheme will comprise of

- Air package
  (Air compressor, Air dryer, Air Receiver, Water Separator)
- Industrial Oxygen generator with oil removal filtration system
- Storage Tank
- Oxygen Analyzer
- Alarm panel

Unique Features of Absstem

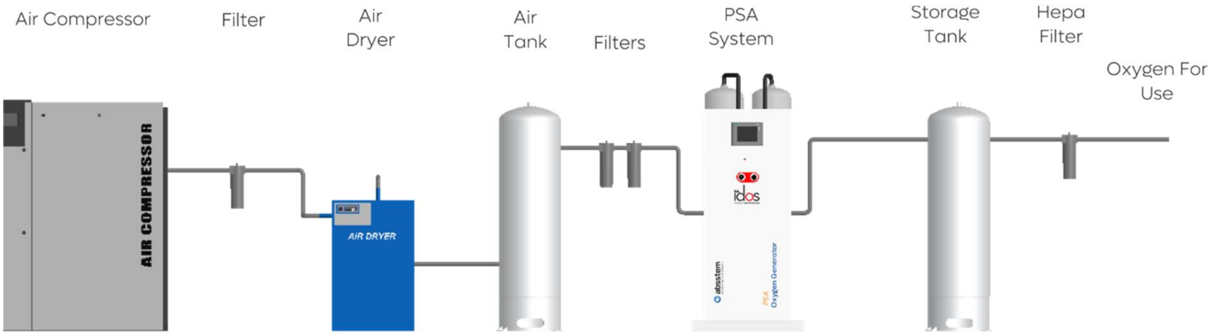
a. All fabricated items are cleaned and degreased as per GEP standards.
b. All Instruments used are compatible with oxygen gas.
c. Guaranteed 10+ years of trouble free and reliable plant operation.
d. We provide the purest **Oxygen** in Industry.
e. Data logging and Graphical Reports for all parameters available.
f. Test certificates of all instruments are available on demand.

Onsite Gas Generation vs Liquid Tank/Cylinder Alternative

Non-stop availability: 24 x 7 across 365 days in an year
Easy integration within existing installations
Your own independent supply source of industrial gas
Significant economies for scale and lower operational costs: no rental charges, logistical expenses and bulk user evaporation losses
No safety hazards related to handling high-pressure cylinders

Equipment Description

The Industrial Oxygen Generator consists of



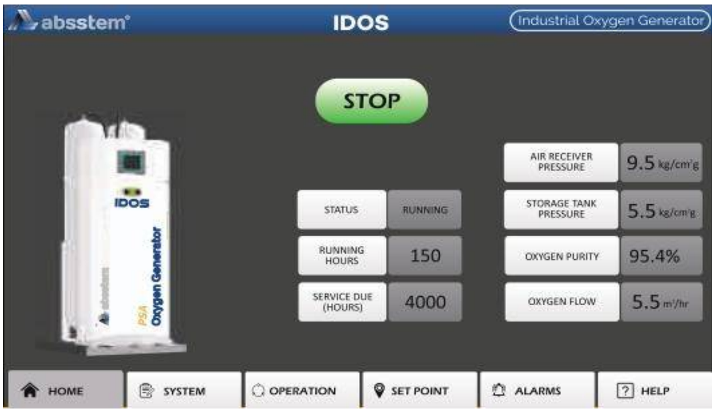
Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **oxygen** from the air.
Refrigerated Air Dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Receiver	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Oxygen** from Air.
Storage Tank	Industrial Grade **Oxygen** produced from IDOS is stored in this tank to supply a continuous flow of **Oxygen**.
Online Oxygen Analyzer	Online Oxygen analyzer for continuous monitoring of Oxygen purity level.

HD Touch Screen Display

(User-friendly touch screen interface for easy operation and maintenance)

Absstem industrial oxygen generator complies with all industry standards and regulation

- Alarm management system
- Plug and play
- Fully automatic
- Process data display
- Consumables running hours display
- Data logging via USB drive
- Graphical representation



STATUS RUNNING
RUNNING HOURS 150
SERVICE DUE (HOURS) 4000
AIR RECEIVER PRESSURE 9.5 kg/cm²g
STORAGE TANK PRESSURE 5.5 kg/cm²g
OXYGEN PURITY 95.4%
OXYGEN FLOW 5.5 m³/hr

Additional upgrades

- Digital flow meter with totaliser
- High-pressure oxygen tank
- Oxygen booster for high pressure requirements
- Cylinder manifold and alarm management system
- Remote monitoring

Plant Operation

- Unit operates 24-hrs a day, 7-days a week throughout the year reliably, automatically & safely

Automation

- The Plant is fully automatic and needs only electrical power for running the Air compressor
- The machine automatically enters into standby mode when the level in the Storage Tank reaches a preset limit. This helps in optimised Air Compressor operation and reduced power consumption.

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial oxygen generators.

TÜV NORD TÜV India Private Ltd ISO 9001
TÜV NORD TÜV NORD CERT INDIA ISO 14001
TÜV NORD TÜV NORD CERT INDIA ISO 13485
RIR CERTIFICATION Partners in Growth

Itemised Supply description

No.	Item description	Make	Qty
a	Air Compressor	Kaeser/IR/As per Demand	Client scope/ Quoted Separately
b	Refrigeration Dryer with pre filter	Kaeser/IR/As per Demand	Client scope/ Part of Compressor Package.
c	Air Tank	ABSSTEM	Client scope/ Part of Compressor Package.
d	Set of Filters	ABSSTEM	1 no.
e	PSA unit (IDOS)	ABSSTEM	1 no.
f	Electrical Power Panel with PLC and cabling	ABSSTEM & MITSUBISHI	1 no.
f	Digital Oxygen Analyzer	ABSSTEM	1 no.
g	Storage Tank	ABSSTEM	Quoted Separately.

Electrical Protection

The client should make proper arrangements for earthing/grounding systems for the Industrial **Oxygen** Generator at their facility.

Enclosure Panel

- Ergonomically designed metal enclosure with powder-coated finish for longer life.
- High-strength IP-rated panel fittings from Germany for a robust enclosure.
- Fan filters from Rittal are IP-55 rated making the electrical enclosure internal environment dust-free.
- Easy connectivity to any place of installation with req. port connections.
- High-strength base frame designed for higher load and easy transportation.
- Vessels are designed per ASME section VIII DIV I. Vessels are epoxy paint coated which is considered good for longevity and corrosion resistance.
- All electrical items are CE certified with operating voltage at 24 VDC

Maintenance Contract (AMC / CMC)

Our in-house service support packages Absstem Shield & Absstem Shield premium ensures 24*7 on site support for all scheduled and need based service requirements. Per the schedule mentioned in the operational manual, we suggest the user to change the filters by following a simple protocol.

Our technical experts are always available to guide your maintenance team for carrying out minor troubleshooting and basic maintenance activities via remote support program.

Best ROI Guaranteed

ROI within 6 to 18 months when compared to conventional methods

Generator performance depends on the component selection. Our processes make this job easier as all equipment is selected based on more than 3 decades of research. IDOS is a fully automatic (operator-free operation) that ensures peace of mind.

We deliver what we commit!!

Product Description

Product	Description	Make	Qty
IDOS Air Package	IDOS 45 CP Oxygen		1

Caption	Specification
Air Compressor KW	55 KW
Air Compressor Make	Atlas Copco/ IR/ Power Efficient Brands
Air Dryer	Included
Inter Connecting Cable	Included
Interconnecting Piping	Included
Pre filter	Included
Dry Air Generated	360 CFM

Product	Description	Make	Qty
IDOS Oxygen Generator	IDOS 45 B Oxygen		1

Caption	Specification
Gas Generated	Oxygen
Flow	45 Nm3/Hr
Purity	95 Percent
Compressor	In Your Scope / Quoted Separately
Dryer	In Your Scope / Part of Compressor Package
Air Vessel	In Your Scope / Part of Compressor Package
Dry Air Required	360 CFM
Power Required	230V 50 Hz Single Phase
Module Option	PSA Plant Only

Product	Description	Make	Qty
IDOS Gas Storage	IDOS 60 ST Oxygen		1

Caption	Specification
Water Volume	13000 Litres
Gas Stored	Oxygen
Gas Stored Volume	71.5 M3
Working Pressure	5.5 Barg
Hydrotest Pressure	9 Barg

Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	GST(%)	Amount
1	IDOS 45 CP	IDOS 45 CP	IDOS Air Package	8414	1	18,18,000.00	18.00	18,18,000.00
2	IDOS 45 B	IDOS 45 B	IDOS Oxygen Generator	842139	1	42,06,000.00	18.00	42,06,000.00
3	IDOS 60 ST	IDOS 60 ST	IDOS Gas Storage	842139	1	8,00,000.00	18.00	8,00,000.00
							Sub Total	68,24,000.00
							GST(%) @18.00	0.00
							Grand Total ()	80,52,320.00

Terms And Conditions

1.	EXCLUSIONS & CLIENT'S SCOPE	All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Client's site will be in Client Scope. Additional gas piping, cooling water (if required).
2.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
3.	P&F	3% Extra.
4.	Freight	Extra at Actuals.
5.	Insurance	In your Scope.
6.	Payment Terms	50% Advance with your Prestegious Purchase Order and Balance against PI.
7.	Delivery	6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability.
8.	Warranty	We hereby confirm that, our equipment shall have Supplier's warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
9.	GST	As per the Commercial Table.
10.	Remarks	.

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For Absstem Industrial Solutions Pvt. Ltd.

Pramod Kumar

Director

8447642325

www.absstem.com